Temporal Fusion Transformer vs. Time Series Transformer for Multi-Asset Cryptocurrency Forecasting

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1 Introduction

The application of artificial intelligence (AI) in finance has been a widely studied topic since its inception. A bibliometric review by Roy et al., 2025 shows research in this field growing since 1993, with an enormous jump from 2017 onward, which can be attributed to the introduction of transformer-based neural networks by Vaswani et al., 2017. The transformer architecture, built on self-attention mechanisms, offered a significant departure from recurrent approaches by enabling parallel sequence processing and the capture of long-range temporal dependencies, properties that proved highly transferable to time series forecasting tasks (Su et al., 2025). Subsequent architectural variants have extended this foundation: the Informer Zhou et al., 2021 introduced used a ProbSparse attention mechanism to reduce the quadratic complexity of standard self-attention for long-horizon tasks, and has since been adopted as the basis for several general-purpose time series forecasting implementations. One domain in which this transferability has been extensively explored is the prediction of price and volume data in financial markets, including both stocks and cryptocurrencies.

Cryptocurrency markets present a particularly challenging forecasting environment. Unlike traditional assets, cryptocurrencies trade continuously, are highly sensitive to sentiment and geopolitical events, and lack the fundamental valuation anchors available to equity analysts. Fama, 1970 establishes the efficient market hypothesis (EMH), under which systematic price prediction would be impossible, yet Urquhart, 2016 demonstrates that Bitcoin exhibits significant market inefficiency, providing empirical justification for the forecasting enterprise. This inefficiency, combined with high volatility and nonlinear dynamics, makes cryptocurrency markets a natural testbed for advanced deep learning architectures.

Trembovetskyi et al., 2024 explore the use of transformer models in stock price and volume forecasting, noting the ability to capture intricate temporal dependencies and the increased accuracy when compared to traditional forecasts. These advantages have been documented across a growing body of literature applying transformers to financial time series (Izadi and Hajizadeh, 2025; Montenegro, 2025; Tokajuk and Chudziak, 2025). Within this broader class of architectures, the Temporal Fusion Transformer (TFT) introduced by Lim et al., 2021 and the Time Series Transformer (TST) based on the Informer architecture from Zhou et al., 2021 have attracted particular attention: TFT for its combination of multi-horizon forecasting performance with interpretable attention-based outputs, and TST for its purer reliance on self-attention without a recurrent component. This paper directly compares these two architectures for cryptocurrency price prediction, motivated by the observed limitations of existing approaches in feature integration, evaluation scope, and methodological rigor. The following sections review the relevant literature and outline the experimental design adopted to address these gaps.

2 Literature Review

2.1 Transformers in Financial and Stock Market Forecasting

The application of transformer architectures to financial time series has produced encouraging results across asset classes. Wu et al., 2020 demonstrate that deep transformer models can capture complex temporal patterns in epidemiological time series, an early example of the architecture's transferability to non-linguistic sequential data with irregular dynamics. In a financial context, Trembovetskyi et al., 2024 evaluate transformer models on stock price and trading volume data, finding that the architecture captures short- and long-term dependencies more effectively than conventional statistical and recurrent baselines, and that predictive accuracy improves relative to traditional forecasting methods. Izadi and Hajizadeh, 2025 combine a transformer encoder with parallel convolutional neural network blocks in a structure inspired by the Inception model. Evaluated on three cryptocurrency datasets, the model achieves 57% directional accuracy on Bitcoin test data, outperforming LSTM and CNN baselines in trend classification tasks, though the binary framing of prediction as a classification problem limits direct comparison with regression-based forecasting approaches.

2.2 Transformer-Based Cryptocurrency Price Forecasting

Several studies have applied transformer architectures directly to cryptocurrency price prediction. Montenegro, 2025 train and evaluate models combining transformer encoders with

LSTM and GRU layers on Bitcoin and Ethereum price data spanning 2017 to 2024, finding that the hybrid architectures produce competitive regression performance relative to equivalent studies in the literature. Tokajuk and Chudziak, 2025 propose a Partial Multivariate Transformer that applies feature selection prior to the attention mechanism, which reduces the influence of noisy or irrelevant covariates on prediction. Evaluated on cryptocurrency time series, the approach shows that selective multivariate input can improve forecasting accuracy over full-covariate transformer baselines. Song et al., 2025 incorporate the Crypto Fear and Greed Index as an external input to an enhanced transformer model for Bitcoin price prediction, combining seasonal decomposition with sentiment-derived signals to capture both structural and behavioral components of price dynamics. The model outperforms standard transformer baselines across multiple datasets, suggesting that external indicators carry predictive information beyond what is contained in price history alone.

2.3 Temporal Fusion Transformer Applications in Cryptocurrency Forecasting

The Temporal Fusion Transformer, introduced by Lim et al., 2021, combines recurrent layers for local processing with interpretable self-attention for long-range dependencies, and incorporates gating mechanisms to suppress uninformative inputs and variable selection networks to identify relevant features across static, known-future, and observed-past input categories. This architecture has been applied to cryptocurrency forecasting in several recent studies, each with notable strengths and limitations.

Farooq et al., 2024 develop an Advanced Deep Learning-Enhanced TFT (ADE-TFT) for Bitcoin price forecasting, incorporating market sentiment analysis and geopolitical data alongside price history. The model outperforms recurrent baselines across MAPE, MSE, and RMSE metrics, with the best configuration using eight hidden layers, and the authors highlight the sensitivity of results to normalisation strategy and input feature selection. However, the evaluation is conducted on a single asset and the contribution of individual input modalities is not isolated, making it difficult to assess which features drive the observed performance gains.

Peik et al., 2025 propose an Adaptive TFT framework for short-term cryptocurrency price forecasting, addressing the limitation of fixed-length windowing by partitioning the volatility rate series into variable-length subseries ending at thresholded relative maxima, then selecting a specialized TFT model based on the binary end-pattern of the preceding subseries. Evaluated on 10-minute ETH-USDT data, the framework outperforms standard LSTM, TFT, and the authors' prior fixed-length baseline in both directional accuracy (51.36%) and simulated trading profitability. However, the evaluation covers a single asset over a one-week test period characterized by abnormal upward volatility, and model gains are driven primarily by high recall (92.31%) at the cost of very low specificity (19.28%), suggesting performance is strongly regime-dependent. The framework also relies exclusively on price-derived volatility, excluding volume, technical indicators, and sentiment signals.

Lee, 2025 proposes a TFT-based forecasting framework for multi-asset cryptocurrency trading, combining technical indicators (RSI, MACD) with on-chain metrics (SOPR, TVL, AA, ENF, HODL Waves) and the Crypto Fear and Greed Index. Evaluated on BTC, ETH, USDT, XRP, and BNB over 2022 to 2024, the model outperforms LSTM, GRU, XGBoost, and SVR baselines with an average MAPE of 3.18% and R^2 of 0.9432, while a derived signal-based trading strategy achieves a cumulative return of 38.6% and Sharpe ratio of 1.06. Despite these results, the study carries notable methodological limitations. Technical and on-chain indicators are treated as known future covariates within the TFT architecture, yet price-derived features such as RSI and MACD require the closing price being predicted, introducing data leakage. Additionally, the absence of a reported validation split leaves the early stopping procedure unspecified. These limitations inform the stricter temporal feature classification and experimental design adopted in the present work.

2.4 Synthesis and Research Gap

Across the reviewed literature, several common patterns emerge. Transformer-based architectures consistently outperform recurrent and statistical baselines in cryptocurrency forecasting tasks, with performance gains attributed to the ability to model long-range dependencies and selectively weight input features (Farooq et al., 2024; Lee, 2025; Tokajuk and Chudziak, 2025). The integration of external signals, including sentiment indices and on-chain metrics, further improves predictive performance over price-history-only models (Lee, 2025; Song et al., 2025). However, a number of limitations recur across studies. Evaluations are frequently restricted to a single asset or a short test window, limiting the generalizability of reported results (Farooq et al., 2024; Peik et al., 2025). Methodological issues including data leakage through incorrect covariate classification and underspecified training procedures introduce uncertainty about whether reported gains reflect genuine predictive capacity (Lee, 2025). No existing study directly compares TFT against a pure attention-based TST architecture under controlled conditions across a large, heterogeneous asset panel. The present work addresses these gaps by adopting a multi-asset evaluation scope, applying strict temporal separation of input features to avoid leakage, and following a fully specified training and validation procedure.

3 Data & Methodology

This section describes the dataset construction pipeline, the feature engineering process, and the two transformer-based architectures used for multi-step cryptocurrency price forecasting. Both models are trained on the same data and evaluated on the same held-out validation window to ensure a fair comparison.

3.1 Data

Price and volume data for 373 cryptocurrency pairs denominated in USDT were retrieved from the Binance API[1] at daily resolution, spanning from August 2017 to March 2026. Each ticker was downloaded independently and concatenated into a flat panel DataFrame indexed by `(timestamp, tic)`. Memecoins were excluded to prevent speculative micro-caps from distorting the results.

Missing trading days were identified per ticker and imputed via linear interpolation between the nearest available neighbors within a seven-day window, with a small amount of Gaussian noise ($\sigma = 0.1 \times$ local standard deviation) added to prevent exact replication artifacts.

Of the 373 tickers, 260 are retained for model training and evaluation after applying two filters.

1. tickers with fewer than 374 training days prior to the cutoff were removed, ensuring each asset can form at least one complete training sample with a full encoder warm-up.
2. tickers present exclusively in the validation window, assets listed after September 2025, were removed, as the TFT architecture requires a learned static embedding per ticker that cannot be constructed without prior training observations. The full list of retained tickers with their start dates and observation counts is provided in Table 4 and Table 5 in the Appendix.

The training set covers 17 August 2017 to 31 August 2025, and the validation set covers 1 September 2025 to 15 March 2026, yielding 342,614 and 47,580 ticker-day observations respectively. Each validation sample uses a 90-day encoder context, meaning the first validation forecast draws on observations from as early as 3 June 2025.

3.1.1 Technical Indicators

Following the feature engineering approach of Gort et al. (2022), nineteen technical indicators were computed for each ticker using TA-Lib[2]. These can be grouped into four categories.

Momentum indicators

- **RSI** (14-period): measures the speed and magnitude of price changes, bounded in $[0, 100]$: RSI $= 100 - \frac{100}{1+\frac{\text{avg gain}}{\text{avg loss}}}$
- **MACD** (12/26/9): difference between a fast and slow exponential moving average, used to identify trend reversals: MACD $= \text{EMA}_{12} - \text{EMA}_{26}$
- **CCI** (20-period): measures deviation of price from its statistical mean, sensitive to overbought/oversold conditions: CCI $= \frac{p_t - \bar{p}}{0.015 \cdot \text{MAD}}$, where p_t is the typical price and MAD is the mean absolute deviation.
- **DX** (14-period): directional movement index, capturing the strength of a trend regardless of direction.
- **ROC** (20-period): percentage change in price over a fixed lookback window: ROC $= \frac{p_t - p_{t-20}}{p_{t-20}} \times 100$
- **ULTOSC**: combines momentum across three timeframes (7, 14, 28 periods) to reduce false divergence signals.
- **WILLR**: Williams %R, a bounded oscillator in $[-100, 0]$ measuring the closing price relative to the high–low range over 14 periods.

Volume indicators

- **OBV**: On-Balance Volume, a cumulative sum of volume signed by daily price direction, used as a leading indicator of price breakouts.
- **HT_DCPHASE**: Hilbert Transform Dominant Cycle Phase, which decomposes price into an instantaneous phase signal to identify cyclical turning points.

Volatility indicators

- **ATR** (14-period): Average True Range, measuring the average of the true range $\max(H - L, |H - C_{t-1}|, |L - C_{t-1}|)$ over the lookback window.
- **NATR** (14-period): ATR normalised by the closing price, enabling volatility comparison across tickers with different price scales: NATR $= \frac{\text{ATR}}{C_t} \times 100$
- **BB_width** (20-period): Bollinger Band width, capturing the relative width of the price envelope: BB_width $= \frac{BB_{upper} - BB_{lower}}{BB_{mid}}$

Price structure indicators

- **EMA cross**: difference between the 21-period and 50-period EMA, where positive values indicate a bullish regime: EMA_cross $= \text{EMA}_{21} - \text{EMA}_{50}$
- **Candle body**: signed body size, capturing both direction and magnitude of the daily move: candle_body = close − open
- **Upper wick**: length of the upper rejection shadow, indicating selling pressure at intraday highs: upper_wick = high − max(open, close)
- **Lower wick**: length of the lower rejection shadow, indicating buying pressure at intraday lows: lower_wick = min(open, close) − low

Features with a Pearson correlation coefficient exceeding 0.9 were removed prior to training to reduce redundancy in the input space. The raw `open`, `high`, and `low` columns were dropped after indicator computation, as their information is implicitly encoded in the derived features above.

In addition, we created a **crypto market sentiment index**, that is daily market-wide score derived from the Augmento.ai API[3], which aggregates social-media sentiment from Twitter, Reddit, and Bitcointalk. For each of N assets we collect the

[1] https://www.binance.com/en/binance-api

[2] https://ta-lib.org/

[3] https://augmento.ai

daily combined sentiment score $s_{i,t}$ and compute a market-capitalization-weighted average:

$$\text{SentIdx}_t = \frac{\sum_{i=1}^{N} \text{MCap}_i \cdot s_{i,t}}{\sum_{i=1}^{N} \text{MCap}_i} \tag{1}$$

where MCap_i is retrieved from Yahoo Finance.

3.1.2 Market Indicators

To capture macroeconomic and cross-asset uncertainty, i.e. factors that technical indicators derived solely from price cannot represent, we added sixteen market-wide indicators were collected from three external sources and merged into the panel by date.

Equity volatility the CBOE Volatility Index (VIX), realised volatility of VIX over a 20-day rolling window (RVOL_VIX), and the VIX term structure slope ($\text{VIX}_{3M} - \text{VIX}$), which captures the shape of the implied volatility curve.

Credit risk one-day and five-day differences of the ICE BofA High-Yield Option-Adjusted Spread ($\text{HY\_OAS}_{\Delta 1}$, $\text{HY\_OAS}_{\Delta 5}$) and the credit risk appetite delta (HY_OAS − IG_OAS), all sourced from the Federal Reserve Economic Data (FRED) database[4].

Macro and rates the 10Y–2Y Treasury yield spread (T10Y2Y), the 10Y–3M spread (T10Y3M), a binary yield curve inversion flag, the TIPS 10-year breakeven inflation rate, and the DXY U.S. Dollar Index with its one-day and five-day differences.

Cross-asset spot gold price (GOLD), the CBOE Crude Oil Volatility Index (OVX), and the Crypto Fear & Greed Index[5], normalised to $[0, 1]$.

Geopolitical risk the Geopolitical Risk Threat Index (GPRT) from Caldara and Iacoviello (2022), extracted from the daily XLS file published at matteoiacoviello.com. The threat sub-index was preferred over the headline GPR index as it captures *anticipated* geopolitical risk, which has greater predictive value than the act sub-index, which reflects events that have already materialised.

Monetary policy the Federal Funds futures surprise signal, computed as FF_surprise = (100 − ZQ close) − DFF, where ZQ is the front-month 30-day Fed Funds futures contract from Yahoo Finance and DFF is the daily effective Federal Funds rate from FRED. A positive value indicates that the market expects rate hikes relative to the current rate.

All market indicators are lagged by one trading day prior to merging with the price panel to eliminate any forward-looking bias. Weekend and public-holiday gaps in macro data are forward-filled, reflecting the realistic condition that a practitioner would use the most recently available value. In Table 1 the list of indicators used in the models.

Table 1: Input features used by both models. Calendar features are known in the future window; all other features are zeroed in TST's `future_time_features` to prevent leakage. TFT encodes calendar features as learned embeddings via `time_varying_known_categoricals`.

Group	Feature(s)
Price & Volume (rolling z-score)	
	close, volume, OBV
Technical Indicators (16 features)	
Momentum	RSI, MACD, CCI, DX, ROC, ULTOSC, WILLR
Volume	HT_DCPHASE
Volatility	ATR, NATR, BB_width
Structure	EMA_cross, candle body, upper/lower wick
Sentiment	Sentiment index
Market Indicators (16 features)	
Eq. vol.	VIX, RVOL_VIX, VIX term slope
Credit	HY_OAS Δ_1, Δ_5, HY–IG spread
Macro	DXY, DXY Δ_1/Δ_5, T10Y2Y, T10Y3M
	Yield curve inv. flag, GOLD
Crypto	Fear & Greed index
Geopolitical	GPRT (threat sub-index)
Mon. policy	FF surprise
Calendar Features - Known Future	
	Day of week (0–6)
	Month (1–12)
	Is weekend (0/1)
Total features	**38**

3.1.3 Scaling

Price-derived features (`close`, `volume`, `OBV`) are normalised using a per-ticker rolling z-score with a 90-day window:

$$\tilde{x}_t = \frac{x_t - \mu_{t-90:t-1}}{\sigma_{t-90:t-1} + \varepsilon}, \tag{2}$$

where $\varepsilon = 10^{-8}$ prevents division by zero. The rolling statistics μ and σ are computed from a shifted window (lag 1) so that the normalisation at time t uses no information beyond $t-1$. The first 90 rows per ticker are discarded as warm-up. The scale parameters (μ, σ) are stored alongside the data and used at inference time to invert predictions back to USD.

Technical indicators are forward-filled and zero-imputed without scaling, as they are already dimensionless or on bounded scales by construction.

Market indicators are scaled using `RobustScaler` (median/IQR) for heavy-tailed series (VIX, RVOL_VIX, GPRT, HY spreads, GOLD) and `StandardScaler` for mean-reverting series (DXY, yield spreads, VIX term slope, FF_surprise). The Crypto Fear & Greed Index and the yield curve inversion flag are passed through without scaling, as they are already bounded in $[0, 1]$ and $\{0, 1\}$ respectively. All scalers are fitted exclusively on training data (pre-2025) and

[4] https://fred.stlouisfed.org/

[5] https://alternative.me/crypto/fear-and-greed-index/

applied to the full panel, including the validation window, to prevent data leakage.

3.2 Models

Both models receive the same preprocessed panel. The forecasting task is defined as: given a context window of $T_c = 90$ daily observations, predict the next $T_p = 7$ days of scaled closing price for each ticker.

3.2.1 Temporal Fusion Transformer

The Temporal Fusion Transformer (Lim et al., 2021) is a hybrid architecture that combines gated recurrent units for local temporal processing with a multi-head attention mechanism for long-range dependency modelling. Its key components are:

- variable selection networks that learn input importance weights per time step, enabling automatic feature selection;
- a sequence-to-sequence LSTM encoder–decoder that captures short-term dynamics;
- a temporal self-attention layer that attends over the full encoder output;
- quantile output heads that produce probabilistic forecasts at the $\{10\%, 50\%, 90\%\}$ quantiles;

In this work, all 35 features, i.e. technical indicators, sentiment, and market indicators, are registered as `time_varying_unknown_reals`, since neither price-derived nor macro features are observable in the prediction horizon. The ticker identifier is encoded as a static categorical embedding. The model is trained end-to-end using the quantile loss.

$$\mathcal{L}_q = \frac{1}{N}\sum_i \left[q \cdot (y_i - \hat{y}_i)^+ + (1-q)\cdot(\hat{y}_i - y_i)^+\right], \quad (3)$$

summed over $q \in \{0.1, 0.5, 0.9\}$.

3.2.2 Time Series Transformer

The Time Series Transformer (TST) is the encoder–decoder architecture introduced in Zhou et al. (2021) and made available through the HuggingFace `transformers` library Wolf et al., 2020. Unlike TFT, it does not include a gated recurrent component; temporal structure is captured entirely through multi-head self-attention and positional encodings, making it a purer test of the transformer inductive bias for time series.

The model receives `past_time_features` (all covariates over the 90-day context window) and `future_time_features` (like calendar features, day-of-week and month, over the 7-day prediction horizon). All price-derived and macro features are zeroed in the future window to prevent leakage, as their future values would be unknown at inference time. A Student's t distributional head is used for the output, which is well suited to the heavy-tailed return distributions of cryptocurrency markets. Seven lag features ($\ell \in \{1, \ldots, 7\}$) are appended to the input to help the model capture weekly periodicity.

The model is trained using the negative log-likelihood of the Student's t distribution as the loss function, which implicitly learns both location and scale of the predictive distribution.

3.2.3 Hyperparameters

To ensure a fair comparison, both models are configured to approximately equal parameter counts (~2.2M) and trained with the same effective batch size of 1,024 samples. Table 2 summarises the key hyperparameters.

Table 2: Hyperparameter configuration. Effective batch size matched at 1,024 via gradient accumulation in TFT.

Hyperparameter	**TFT**	**TST**
Architecture		
Hidden size / d_{model}	128	128
LSTM layers	3	—
Encoder / Decoder layers	—	5 / 5
Attention heads	8	8
FFN dimension	—	512
Hidden continuous size	64	—
Output head	Quantile	Student t
Quantiles / samples	$\{.1, .5, .9\}$	200
Lag sequence	—	$\{1 \ldots 7\}$
Parameters	2,219,617	2,326,659
Training		
Physical batch size	512	1,024
Grad. accum. steps	2	1
Effective batch size	1,024	1,024
Learning rate	7.5×10^{-4}	1.0×10^{-4}
Optimiser	Adam	AdamW
Weight decay	—	10^{-3}
Gradient clip	0.1	0.1
Dropout	0.3	0.3
Precision	BF16	FP32
Max epochs	100	100
Early stop patience	12	15
LR scheduler	ReduceLROnPlateau (p=5, f=0.7)	
Data		
Context / pred. length	T_c=90, T_p=7	
Scaler window	90 days	
Train / val cutoff	2025-09-01	
Min. training days	374	
Tickers (train & val)	260	

The learning rate for TFT follows the linear scaling rule (Goyal et al., 2017): relative to a reference batch size of 256, the learning rate is scaled proportionally as $\eta = \eta_0 \times (B/256)$. TFT uses BF16 mixed precision to reduce memory pressure on the NVIDIA RTX 5070 Ti GPU, while TST uses full FP32 precision, as its training loop it is more lightweight compared to TFT.

3.3 Evaluation Metrics

To compare the two models, we use six metrics and one statistical test. All point-error metrics (MAE, RMSE, MAPE) are computed in the original USD price domain after inverse-scaling, whereas scale-invariant metrics (MASE, Pinball Loss, Directional Accuracy) are computed in the normalized domain to prevent high-price assets such as Bitcoin from dominating the aggregate scores. All metrics are first averaged per ticker and then across tickers to give equal weight to each asset regardless of its price level.

Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{4}$$

MAE measures the average magnitude of forecast errors in USD and is robust to outliers compared to squared-error alternatives.

Root Mean Squared Error (RMSE)

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{5}$$

RMSE penalizes large errors more heavily than MAE, making it sensitive to occasional large mispredictions, which are common in cryptocurrency markets during high-volatility episodes.

Mean Absolute Percentage Error (MAPE)

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{|y_i|} \tag{6}$$

MAPE expresses forecast error as a fraction of the actual price, enabling comparison of accuracy across assets with different price scales. Values near zero indicate close percentage tracking of the price series.

Mean Absolute Scaled Error (MASE)

$$\mathrm{MASE} = \frac{\mathrm{MAE}_{\mathrm{model}}}{\mathrm{MAE}_{\mathrm{naïve}}} = \frac{\frac{1}{N}\sum_{i=1}^{N}|\tilde{y}_i - \hat{\tilde{y}}_i|}{\frac{1}{N-1}\sum_{i=2}^{N}|\tilde{y}_i - \tilde{y}_{i-1}|} \tag{7}$$

where $\tilde{y}$ denotes scaled prices. MASE normalizes model error by the error of a naive random-walk baseline, defined as the forecast $\hat{y}_{t+1} = y_t$, i.e. tomorrow's price is predicted to be equal to today's price. This baseline is deceptively competitive in financial time series: if prices follow a random walk (Fama, 1970), no model can systematically outperform it, and empirical evidence suggests that cryptocurrency markets exhibit near-random-walk behavior over short horizons as per Urquhart, 2016. The denominator of MASE captures the average one-step absolute change in the scaled price series, which represents the irreducible difficulty of the forecasting task for that particular asset. A MASE value below 1 indicates that the model outperforms the naive baseline on average; a value above 1 indicates it does not. Reporting MASE alongside MAE is therefore essential in our setting: a model may achieve a low absolute MAE simply because the asset it is predicting is low-volatility and slowly drifting, while the naive baseline would perform equally well. MASE corrects for this by anchoring the evaluation to the difficulty of each individual asset.

Prediction Interval Coverage

$$\mathrm{Coverage}_{[10,90]} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\left[p_i^{(10)} \le y_i \le p_i^{(90)}\right] \tag{8}$$

Coverage measures the empirical proportion of actual observations falling within the $[P_{10}, P_{90}]$ prediction interval. A well-calibrated probabilistic forecast should yield coverage close to the nominal 80%.

Directional Accuracy

$$\mathrm{DA} = \frac{1}{N-1}\sum_{i=2}^{N}\mathbf{1}\left[\mathrm{sign}(\tilde{y}_i - \tilde{y}_{i-1}) = \mathrm{sign}(\hat{\tilde{y}}_i - \hat{\tilde{y}}_{i-1})\right] \tag{9}$$

Directional accuracy measures how often the model correctly predicts the sign of the next-day price change, computed exclusively on the $t+1$ horizon to avoid inflation from within-window horizon correlation. This metric is of direct relevance to trading applications where the direction of the move matters more than its magnitude.

Pinball Loss

$$\mathcal{L}_q = \frac{1}{N}\sum_{i=1}^{N}\begin{cases} q\,(y_i - \hat{q}_i) & \text{if } y_i \ge \hat{q}_i \\ (1-q)\,(\hat{q}_i - y_i) & \text{if } y_i < \hat{q}_i \end{cases} \tag{10}$$

The Pinball (quantile) loss is computed at each of the three output quantiles $q \in \{0.1, 0.5, 0.9\}$ and averaged. It directly measures the quality of the probabilistic forecast: a lower Pinball loss indicates better-calibrated and sharper predictive intervals.

Diebold–Mariano Test To assess whether observed differences in forecast accuracy are statistically significant rather than arising by chance, we apply the Diebold–Mariano (DM) test (Diebold & Mariano, 1995). Given the squared forecast errors $e_i^{(a)} = (\tilde{y}_i - \hat{\tilde{y}}_i^{(a)})^2$ and $e_i^{(b)} = (\tilde{y}_i - \hat{\tilde{y}}_i^{(b)})^2$ for models a and b, the loss differential is $d_i = e_i^{(a)} - e_i^{(b)}$, and the DM statistic is:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d})}} \xrightarrow{d} \mathcal{N}(0,1) \tag{11}$$

A negative DM statistic indicates that model a produces lower squared errors than model b. The test is applied in the normalised price domain to give equal weight across all tickers. Statistical significance is assessed at the $\alpha = 0.05$ level.

4 Results

Across all evaluated metrics, the Time Series Transformer (TST) outperformed the Temporal Fusion Transformer (TFT). A summary of the metrics can be found in Table 3.

In terms of point error, TST achieved an MAE of 36.40 USD compared to 73.90 USD for TFT, and a RMSE of 49.74 versus 87.58. Since cryptocurrency has hugely varying prices we used Scale-invariant metrics like MAPE, which produced scores of 12.97% versus 20.36%. MASE followed the same pattern, with TST achieving a MASE of 19.452 against 33.019 and a Pinball Loss of 7.026 against 21.828.

For probabilistic calibration, TST achieved a coverage of 0.574 within the $[P_{10}, P_{90}]$ interval, considerably closer to the nominal 80% target than TFT's coverage of 0.127. Directional accuracy was 0.494 for TST and 0.455 for TFT.

Horizon-specific results show that the performance gap was most pronounced at the short horizon: at $t+1$, TST recorded an MAE of 15.42 USD versus 68.98 USD for TFT, while at $t+7$ the gap narrowed to 55.23 versus 78.70 USD.

Finally, the Diebold–Mariano test yielded a statistic of +242.51 ($p < 0.001$), indicating that the difference in forecast accuracy between the two models is statistically significant.

Table 3: Summary of evaluation metrics. Coverage **Winner (W)** is closest to the nominal 80%. DM $p < 0.001$ indicates TST is significantly more accurate.

Metric	**TFT**	**TST**	**W**
Point error (USD)			
MAE	73.90	**36.40**	TST
RMSE	87.58	**49.74**	TST
MAPE (%)	20.36	**12.97**	TST
Scale-invariant			
MASE	33.019	**19.452**	TST
Pinball Loss	21.828	**7.026**	TST
Probabilistic			
Coverage $[P_{10}, P_{90}]$	0.127	**0.574**	TST
Directional			
Dir. Accuracy	0.455	**0.494**	TST
By horizon (USD)			
$t+1$ MAE	68.98	**15.42**	TST
$t+7$ MAE	78.70	**55.23**	TST
Statistical test			
DM statistic	+242.51	—	TST
DM p-value	<0.001	—	TST

5 Discussion

The results reveal that while the Time Series Transformer consistently outperforms the Temporal Fusion Transformer, neither model surpasses the naive random-walk baseline (MASE > 1). This highlights the near-efficiency of cryptocurrency markets at a daily frequency, where the naive forecast $\hat{y}_{t+1} = y_t$ remains exceptionally robust. The high MASE values (19.45 for TST; 33.02 for TFT) further reflect a low signal-to-noise ratio: in low-volatility periods, the MASE denominator (average daily price change) is so small that even minor prediction errors result in baseline underperformance.

The performance gap between models stems from their architectural philosophies. TST's pure attention mechanism captures abrupt regime shifts more effectively than TFT's hybrid LSTM-attention structure. TFT's recurrent inductive bias, while providing multi-step coherence, acts as a constraint that lags behind sudden market reversals. Consequently, TST achieves superior $t + 1$ responsiveness, whereas TFT exhibits linear but slower-reacting error growth across horizons.

Furthermore, TST's Student's t distributional head is better suited for heavy-tailed returns than TFT's quantile regression. TFT's severe miscalibration (0.127 coverage vs. TST's 0.574) likely arises from the difficulty of jointly satisfying multiple quantile constraints across 260 heterogeneous assets. TST's likelihood-based formulation avoids this multi-objective complexity by learning a continuous predictive distribution, offering a more robust framework for volatile cryptocurrency data.

Several limitations of this study warrant acknowledgment. First, the reliance on daily data precludes the capture of intraday patterns or high-frequency mean reversion. Second, the feature set, while extensive, excludes granular order-book liquidity, advanced on-chain flows, and real-time news sentiment. Third, the fixed train-test split (pre- vs. post-September 2025) lacks the rolling-window retraining necessary to fully simulate a live trading environment. Fourth, hyperparameters were matched for parameter count rather than individually optimized, which may influence the observed performance gap. Finally, while the MASE denominator can be unstable for low-volatility assets, this does not undermine the robust finding that MASE > 1 across diverse tickers.

6 Conclusion

This study compared the Temporal Fusion Transformer and Time Series Transformer for multi-horizon cryptocurrency forecasting across 260 asset pairs. Three primary findings emerge. First, TST significantly outperforms TFT in point accuracy and calibration ($p < 0.001$), particularly at the $t + 1$ horizon (MAE 15.42 vs. 68.98 USD). Second, neither model surpasses the random-walk baseline, with MASE consistently exceeding one for all assets. Third, both models remain miscalibrated, as TST achieved only 0.574 coverage against an 80% target.

For practitioners, these results suggest that while TST is architecturally superior, its directional accuracy (0.494) and prediction intervals are insufficient for risk-sensitive trading without robust position sizing. Future research should prioritize high-frequency data to capture intraday patterns, implement rolling-window retraining for regime adaptation, and explore cross-asset correlations via graph-based architectures. We emphasize that cryptocurrency forecasting studies must report MASE and test against the random-walk baseline to ensure a rigorous evaluation of model utility.

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A Table & Figures

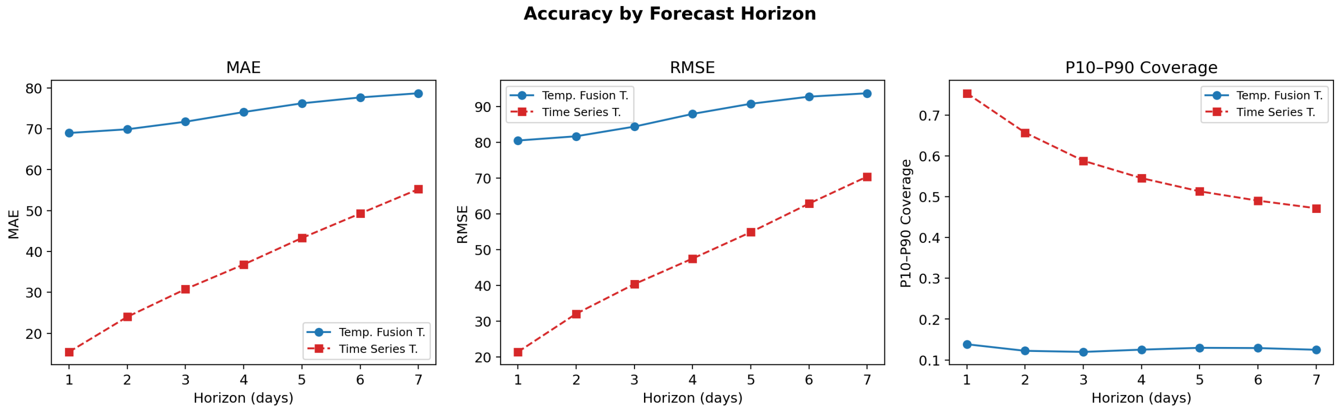

Figure 1: Forecast accuracy by horizon (t+1 to t+7) for MAE, RMSE, and P10–P90 coverage. TFT error grows slowly and near-linearly with horizon, while TST degrades sharply beyond t+2, suggesting the recurrent component in TFT provides better multi-step coherence.

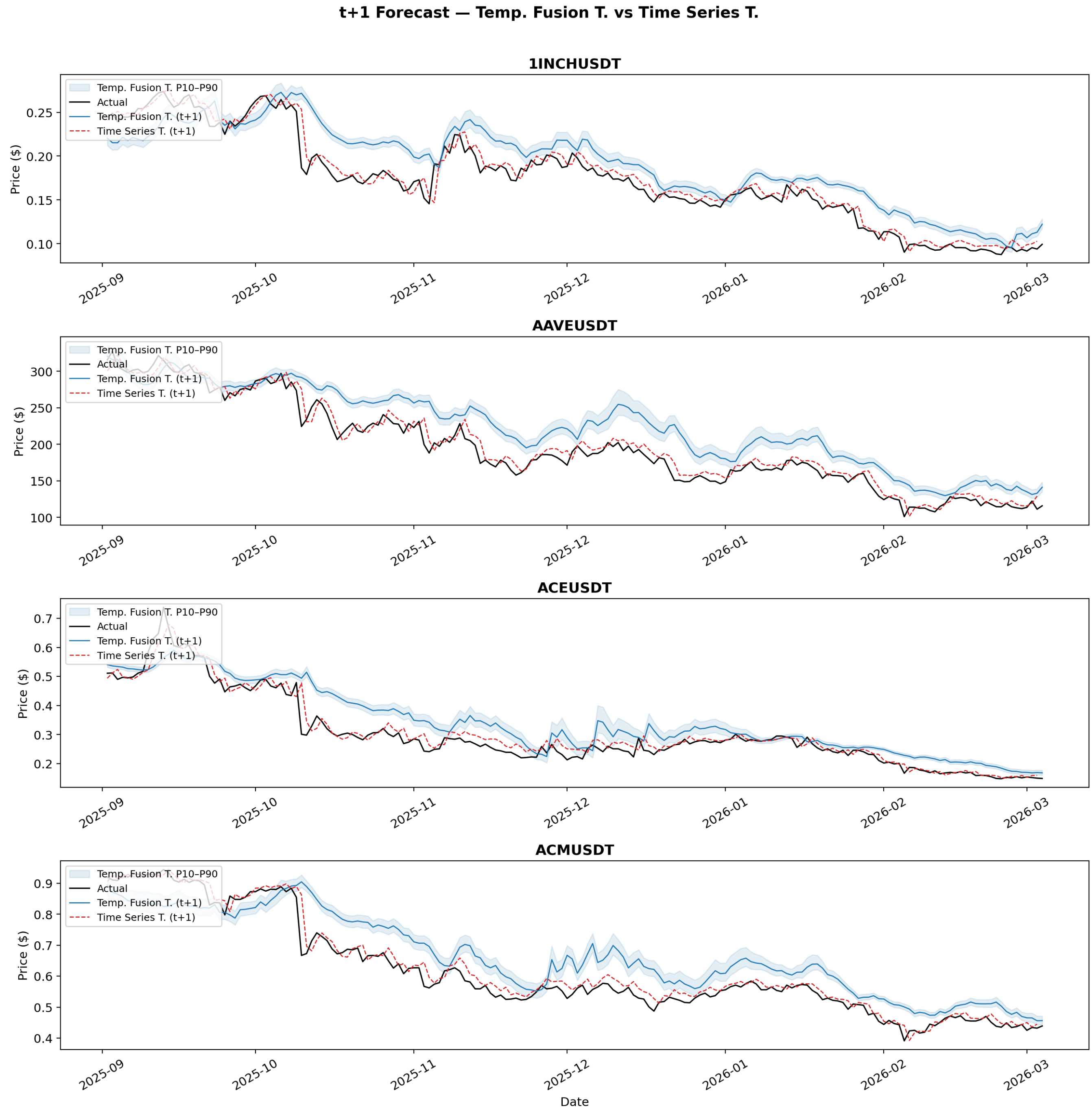

Figure 2: One-step-ahead (t+1) forecasts for four representative assets across the full validation window (January 2025 – March 2026). The shaded band shows TFT's P10–P90 prediction interval. Both models track trend well in calm periods; TST shows larger deviations during sharp reversals (e.g. BNB in November 2025, BTC in early 2026).

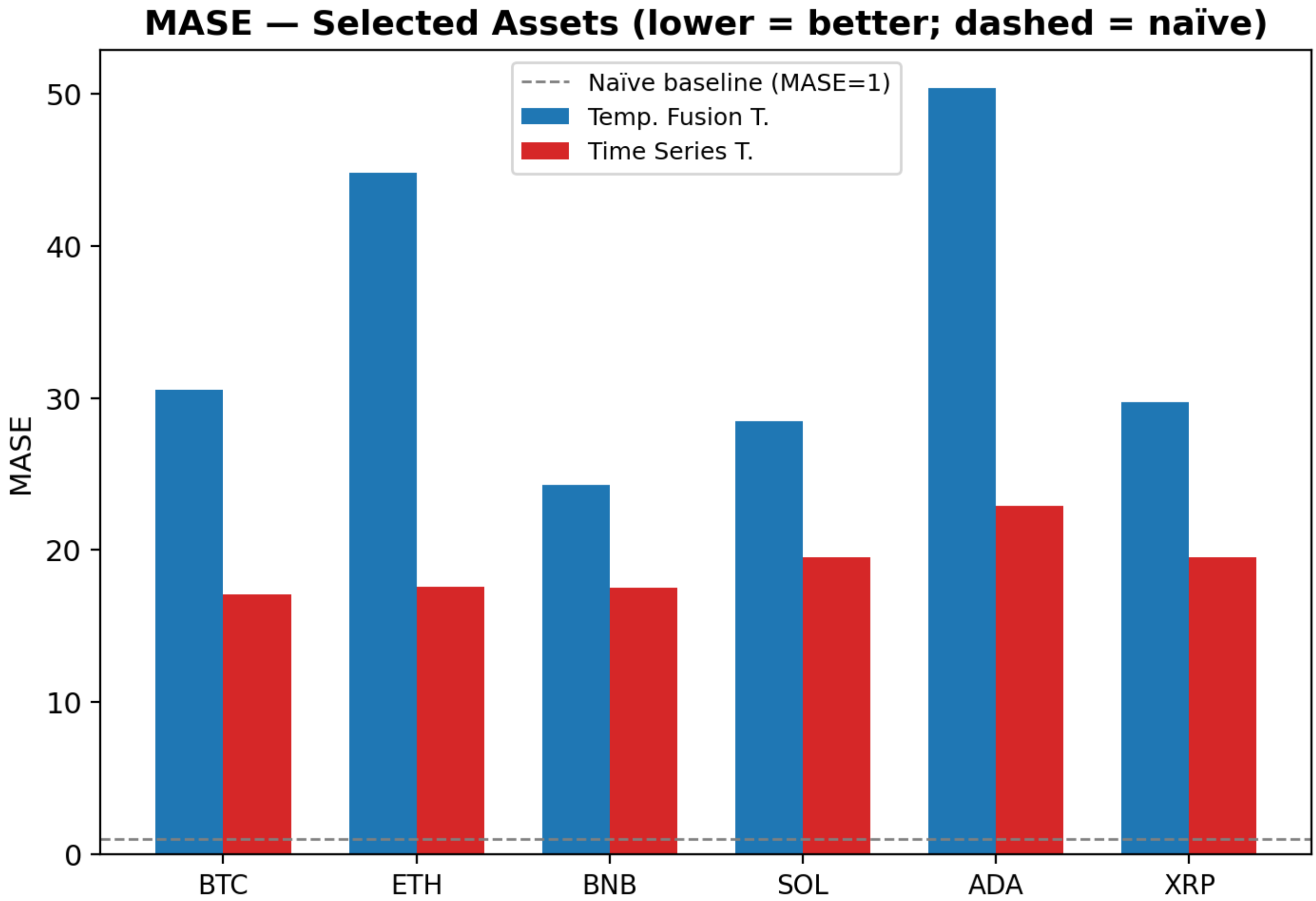

Figure 3: Per-ticker MASE for six select assets in the validation set, namely BTC, ETH, BNB, SOL, ADA, and XRP. The dashed line marks MASE = 1 (naïve random-walk baseline). Both models exceed the baseline on every ticker, indicating that neither architecture beats the random walk on this dataset.

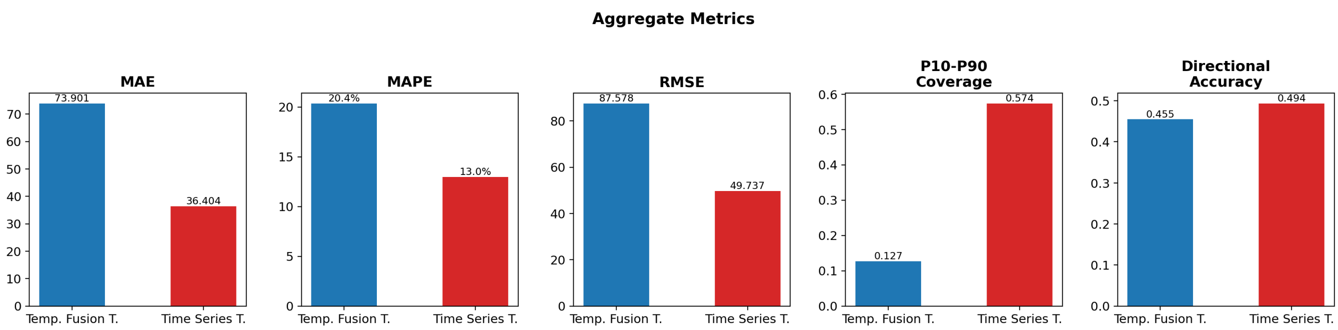

Figure 4: Aggregate metrics averaged across all 16 tickers. TFT dominates on absolute error metrics (MAE, RMSE) and Pinball Loss, while TST achieves better interval coverage and directional accuracy.

Table 4: Retained tickers with first observation date and total trading days (1 of 2). All series end on 2026-03-09.

Ticker	Start	Days	Ticker	Start	Days	Ticker	Start	Days
BTC	2018-01-17	2,974	ARPA	2020-04-07	2,163	RUNE	2021-02-04	1,860
ETH	2018-01-17	2,974	IOTX	2020-04-15	2,155	UMA	2021-02-09	1,855
BNB	2018-04-08	2,893	RLC	2020-04-15	2,155	BEL	2021-02-15	1,849
NEO	2018-04-22	2,879	BCH	2020-04-29	2,141	UNI	2021-02-17	1,847
LTC	2018-05-15	2,856	FTT	2020-05-21	2,119	OXT	2021-02-21	1,843
QTUM	2018-08-19	2,760	EUR	2020-06-04	2,105	AVAX	2021-02-22	1,842
ADA	2018-09-17	2,731	OGN	2020-06-10	2,099	UTK	2021-03-04	1,832
XRP	2018-10-04	2,714	BNT	2020-07-08	2,071	XVS	2021-03-08	1,828
IOTA	2018-10-31	2,687	LSK	2020-07-08	2,071	NEAR	2021-03-16	1,820
TUSD	2018-10-31	2,687	MBL	2020-07-23	2,056	AAVE	2021-03-17	1,819
XLM	2018-10-31	2,687	COTI	2020-07-28	2,051	FIL	2021-03-17	1,819
ONT	2018-11-08	2,679	CTSI	2020-09-23	1,994	INJ	2021-03-23	1,813
TRX	2018-11-11	2,676	HIVE	2020-09-27	1,990	AUDIO	2021-03-25	1,811
ETC	2018-11-12	2,675	CHR	2020-10-07	1,980	CTK	2021-03-29	1,807
ICX	2018-11-13	2,674	ARDR	2020-10-15	1,972	AXS	2021-04-06	1,799
VET	2018-12-25	2,632	MDT	2020-11-05	1,951	STRAX	2021-04-20	1,785
USDC	2019-05-17	2,489	KNC	2020-11-12	1,944	ROSE	2021-04-21	1,784
LINK	2019-06-18	2,457	LRC	2020-11-12	1,944	AVA	2021-04-26	1,779
ONG	2019-07-19	2,426	COMP	2020-11-25	1,931	SKL	2021-05-03	1,772
ZIL	2019-07-22	2,423	ZEN	2020-12-06	1,920	GRT	2021-05-19	1,756
FET	2019-07-31	2,414	SC	2020-12-06	1,920	JUV	2021-05-23	1,752
ZRX	2019-07-31	2,414	SNX	2020-12-09	1,917	PSG	2021-05-23	1,752
BAT	2019-08-04	2,410	VTHO	2020-12-17	1,909	1INCH	2021-05-27	1,748
ZEC	2019-08-21	2,393	DGB	2020-12-20	1,906	OG	2021-06-01	1,743
IOST	2019-08-22	2,392	SXP	2020-12-20	1,906	ASR	2021-06-01	1,743
CELR	2019-08-25	2,389	DCR	2020-12-30	1,896	ATM	2021-06-01	1,743
DASH	2019-08-28	2,386	STORJ	2020-12-30	1,896	CELO	2021-06-07	1,737
THETA	2019-09-10	2,373	MANA	2021-01-06	1,889	RIF	2021-06-09	1,735
ENJ	2019-09-18	2,365	YFI	2021-01-10	1,885	TRU	2021-06-21	1,723
ATOM	2019-09-29	2,354	JST	2021-01-11	1,884	CKB	2021-06-28	1,716
TFUEL	2019-10-24	2,329	SOL	2021-01-11	1,884	TWT	2021-06-29	1,715
ONE	2019-11-01	2,321	SAND	2021-01-14	1,881	SFP	2021-07-11	1,703
ALGO	2019-11-22	2,300	CRV	2021-01-15	1,880	CAKE	2021-07-22	1,692
DUSK	2019-12-22	2,270	DOT	2021-01-18	1,877	DODO	2021-07-22	1,692
ANKR	2019-12-23	2,269	NMR	2021-01-19	1,876	ACM	2021-07-27	1,687
MTL	2020-01-23	2,238	LUNA	2021-01-21	1,874	POND	2021-08-09	1,674
WAN	2020-01-27	2,234	RSR	2021-01-27	1,868	DEGO	2021-08-10	1,673
CHZ	2020-02-06	2,224	PAXG	2021-01-28	1,867	ALICE	2021-08-15	1,668
BAND	2020-02-18	2,212	TRB	2021-01-29	1,866	CFX	2021-08-29	1,654
XTZ	2020-02-24	2,206	SUSHI	2021-02-01	1,863	TKO	2021-09-07	1,645
RVN	2020-02-25	2,205	EGLD	2021-02-03	1,861	PUNDIX	2021-09-09	1,643
HBAR	2020-02-29	2,201	DIA	2021-02-04	1,860	TLM	2021-09-13	1,639
KAVA	2020-03-26	2,175	FIO	2021-02-04	1,860			
STX	2020-03-26	2,175	KSM	2021-02-04	1,860			

Table 5: Retained tickers with first observation date and total trading days (2 of 2). All series end on 2026-03-09.

Ticker	Start	Days	Ticker	Start	Days	Ticker	Start	Days
BAR	2021-09-21	1,631	AMP	2022-04-25	1,415	ID	2023-08-22	931
FORTH	2021-09-23	1,629	PYR	2022-04-28	1,412	ARB	2023-08-23	930
ICP	2021-10-11	1,611	ALCX	2022-05-02	1,408	RDNT	2023-08-30	923
AR	2021-10-14	1,608	SANTOS	2022-05-03	1,407	EDU	2023-09-28	894
MASK	2021-10-25	1,597	BICO	2022-05-11	1,399	WBTC	2023-09-28	894
LPT	2021-10-28	1,594	FLUX	2022-05-12	1,398	SUI	2023-10-03	889
ATA	2021-11-07	1,584	HIGH	2022-05-19	1,391	MAV	2023-11-28	833
GTC	2021-11-10	1,581	CVX	2022-05-25	1,385	PENDLE	2023-12-03	828
PHA	2021-11-25	1,566	JOE	2022-05-30	1,380	ARKM	2023-12-18	813
MLN	2021-12-05	1,556	IMX	2022-06-12	1,367	WBETH	2023-12-19	812
C98	2021-12-23	1,538	GLMR	2022-06-13	1,366	WLD	2023-12-24	807
DEXE	2021-12-23	1,538	API3	2022-06-23	1,356	FDUSD	2023-12-26	805
QNT	2021-12-29	1,532	SCRT	2022-06-23	1,356	CYBER	2024-01-15	785
FLOW	2021-12-30	1,531	BTTC	2022-06-27	1,352	SEI	2024-01-15	785
RAY	2022-01-10	1,520	XNO	2022-06-30	1,349	ARK	2024-02-22	747
MINA	2022-01-10	1,520	WOO	2022-07-11	1,338	IQ	2024-02-22	747
FARM	2022-01-11	1,519	ALPINE	2022-07-24	1,325	NTRN	2024-03-11	729
QUICK	2022-01-13	1,517	T	2022-07-28	1,321	TIA	2024-04-01	708
MBOX	2022-01-19	1,511	ASTR	2022-07-31	1,318	ORDI	2024-04-08	701
REQ	2022-01-20	1,510	GMT	2022-08-09	1,309	BEAMX	2024-04-15	694
WAXP	2022-01-23	1,507	APE	2022-08-17	1,301	PIVX	2024-04-17	692
GNO	2022-01-30	1,500	BIFI	2022-09-01	1,286	BLUR	2024-04-25	684
XEC	2022-02-03	1,496	STEEM	2022-09-22	1,265	VIC	2024-04-25	684
DYDX	2022-02-09	1,490	NEXO	2022-09-29	1,258	VANRY	2024-05-02	677
IDEX	2022-02-09	1,490	LDO	2022-10-09	1,248	AEUR	2024-05-07	672
USDP	2022-02-10	1,489	OP	2022-11-01	1,225	JTO	2024-05-08	671
GALA	2022-02-13	1,486	STG	2023-01-19	1,146	ACE	2024-05-19	660
ILV	2022-02-22	1,477	GMX	2023-03-07	1,099	NFP	2024-05-28	651
YGG	2022-02-24	1,475	POLYX	2023-03-19	1,087	AI	2024-06-05	643
SYS	2022-02-24	1,475	APT	2023-03-21	1,085	XAI	2024-06-10	638
FIDA	2022-03-02	1,469	OSMO	2023-03-30	1,076	MANTA	2024-06-19	629
AGLD	2022-03-07	1,464	HFT	2023-04-09	1,066	ALT	2024-06-26	622
RAD	2022-03-09	1,462	PHB	2023-04-20	1,055	PYTH	2024-07-04	614
RARE	2022-03-13	1,458	HOOK	2023-05-03	1,042	RONIN	2024-07-07	611
LAZIO	2022-03-23	1,448	MAGIC	2023-05-14	1,031	DYM	2024-07-08	610
ADX	2022-03-31	1,440	RPL	2023-06-20	994	PIXEL	2024-07-21	597
AUCTION	2022-03-31	1,440	GNS	2023-07-20	964	STRK	2024-07-22	596
MOVR	2022-04-10	1,430	SYN	2023-07-25	959	PORTAL	2024-07-31	587
CITY	2022-04-12	1,428	SSV	2023-07-27	957	AXL	2024-08-01	586
ENS	2022-04-12	1,428	LQTY	2023-07-31	953	METIS	2024-08-11	576
QI	2022-04-17	1,423	GAS	2023-08-17	936	AEVO	2024-08-13	574
PORTO	2022-04-18	1,422	GLM	2023-08-17	936	ETHFI	2024-08-18	569
POWR	2022-04-19	1,421	PROM	2023-08-17	936			
JASMY	2022-04-24	1,416	QKC	2023-08-17	936			